

Meeting Summaries of July 9

Date: July 9, 2026

Time: 8:00 PM - 9:00 PM Vancouver

Attendees: Feng Gu, Prof. Jan, Prof. Javed

Topic: Data validation, PCA methodology, SUMREL calculation, and preparation of updated thesis-analysis results.

Checklist of Accomplishments since Last Meeting

- On slides page-9, it is added a step from orthogonal loading vectors to the orthogonality among principal component vectors. Meanwhile, the readability and consistency in notations have been improved.
- A new and simpler methodology is presented to argue why the conventional 20% cut-off for least-contaminated sites is not sufficient to define the reference condition. It covers the motivation and resolution of adjusting the cut-off value based on a hypothesis test scenario.
- A detailed explanation is provided on how the log-transformation and standardization (z-score) is influenced by and gets rid of the unit effect. These tangled relationships are clarified in both PC loading $\mathbf{L}$ and PC score $\mathbf{M}'\mathbf{L}$ computations. On top of the explanation, the two approaches that take original chemical matrix and output stressor (principal component) values are compared:

$$f_a : \mathbf{M} \rightarrow \log(\mathbf{M}) \rightarrow z(\log(\mathbf{M})) \rightarrow z(\log(\mathbf{M}))\mathbf{L}$$

$$f_b : \mathbf{M} \rightarrow \log(\mathbf{M}) \rightarrow \log(\mathbf{M})\mathbf{L} \rightarrow z(\log(\mathbf{M})\mathbf{L})$$

Feng chose f_a method and compared it with f_b method that was used in Jian's work.

Questions, Feedbacks and Suggestions

- From original chemical matrix $\mathbf{M}$ to stressor (principal component) matrix $\mathbf{S}$, approach f_b makes more sense. It produce stressor measures that are more harmonized with ecological community response. Additionally, the over earlier standardization of raw variable in f_a approach implicitly causes the stressor measures not factually reflect the relative chemical concentration levels of sites.
- Taking additive view that summarizes the stressor values into a single SUMREL value is a candidate way to define overall contamination level. However, a contamination measure should be scaled into a range start from 0 to some upper limit and the 0 value

needs to be defined as the least-contaminated sites with more caution. On the resulted contamination measure axis, assuming a recorded measure of $a, a \neq 0$, then what does $a + 0 = a$ mean on the contamination measure axis? A site with a record value a does not move its place on the measurement axis but if the 0 value does not mean null contamination, then the $a + 0$ actually moves in a behind place to point to a different contaminated status from original a value.

- The normality in SUMREL in Feng’s results comes from the early standardization of raw variables and the additive view that summarizes the retained 4 stressor values into a single SUMREL value. Specifically, a site t has its 4 stressor values $\mathbf{S}_t = [s_1, s_2, s_3, s_4]$ and each stressor value is a linear combination of the 16 log-transformed and standardized raw variables $[x_1, x_2, \dots, x_{16}]$. Therefore, adding the four stressor values is equivalent to adding the 16 log-transformed and standardized raw variables with weights in each stressor value and the weights are different for each stressor value. Then, the SUMREL value for a site t can be expressed as below:

$$SumRel_t = \sum_{i=1}^4 s_i = \sum_{i=1}^4 \sum_{j=1}^{16} w_{ij} x_j, \quad x_j \sim \mathcal{N}(0, 1)$$

The w_{ij} are loading values in the i -th stressor value for the j -th raw variable. As the x_j are standardized, they all have mean 0 that does get affected by the weights w_{ij} . Consequently, the SUMREL value will be highly concentrated around 0 and the distribution of SUMREL values will be normal-like. In addition, the high correlation among variables with $\rho_{ij} > 0.8$ also contributes to the normality as they can be viewed as drawn from a same distribution most of the time.

Therefore, the mixture normality in this SUMREL value is from adding the 16 standardized raw variables with different weights in each of the 4 stressor values. The 4 stressor values are then added together to form the SUMREL value, and it finally equals to adding 64 weighted standardized raw variables that some groups of them are highly correlated and all of them are 0-mean standardized (not standard variation of 1 due to the weighting).

- Feng presented progress on chemical-data processing, stressor-value construction, and PCA-based contamination analysis.
- Prof. Jan confirmed that the updated chemical-data files sent to Feng use consistent units and follow Jane’s earlier work more closely.
- The group discussed inconsistencies among three reports produced across roughly 10 years, including missing or imputed values and the importance of using the complete 1991 data set.
- The advisors clarified differences between Feng’s approach and Jane’s approach, especially the timing of log transformation, standardization, and removal of unit effects.
- Prof. Jan advised that SUMREL should include PC1, PC2, PC3, PC4, and PC5, while excluding PC6 because it mainly represents non-pollutant or natural elements.
- Prof. Jan recommended interpreting contamination thresholds using ecological meaning and expert judgement, rather than relying only on hypothetical statistical distributions.

Action Items

- Feng will compare the new data set from Prof. Jan with his current data set and document any differences.
- Feng will rerun PCA using the validated data set and compare the results with Jane's PCA.
- Feng will update the PC table and contamination analysis by including PC1 through PC5, excluding PC6, and checking whether PC5 should be flipped for interpretation.
- Feng will implement SUMREL by scaling each selected PC score from 0 to 1 and summing the five scaled scores, giving a 0 to 5 range.
- Feng will review the file from Prof. Jan describing Jane's ordination procedure, including scaling, summing, and selecting the top and bottom 20% of sites.
- Feng will prepare updated figures and results from the revised data and methodology for the next meeting.
- Feng will send static, version-labeled PDFs of thesis chapters to Prof. Jan and Prof. Javed for review.
- Prof. Jan will provide point-form written feedback summarizing Jane's procedure.
- Feng, Prof. Jan, and Prof. Javed will meet again on Monday at 7:00 PM Vancouver time to discuss the updated results.

Summary of Discussion

- The meeting focused on Feng's progress in data validation and PCA-based contamination analysis for the thesis.
- Feng described recent fixes to the chemical data and discussed how large values, missing values, and unit inconsistencies affect stressor construction and PCA results.
- Prof. Jan explained that the updated data files were intended to match Jane's earlier analysis more closely, including consistent units and imputed missing values. Feng expressed confidence that he could reproduce the earlier results once the validated data were used.
- The group discussed the methodological difference between weighting variables equally and weighting principal components equally. Prof. Jan emphasized that each selected PC should be treated as a separate pollution stressor in the SUMREL calculation.
- The team agreed to revise the analysis using PC1 through PC5, exclude PC6, and focus the next meeting on the new results, contamination thresholds, and ecological interpretation.
- The advisors also recommended sending static chapter PDFs for review, because continuously updated documents can make feedback difficult to track.

Action Items

-
- Feng needs to fix these mentioned issues, bring the fixes and prepare the next propotion for the next meeting.

Summary of Discussion

- Just normal report as usual.

Next Meeting Time

- **Date and Time:** July 13, 2026, 7:00 - 8:00 Vancouver
- **Form:** Online, zoom meeting.